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Concept Study of the Heterodyne Spectroscopy Instrument (HSI) for the proposed Far-IR Spectroscopy Space Telescope (FIRSST)

Martina C. Wiedner, Andrey Baryshev, Paul Grimes, Victor Belitsky, Vincent Desmaris, Yan Delorme, Juan-Daniel Gallego, Cristina García-Miró, Uma Gorti, Paul Hartogh, Netty Honingh, Bernd Klein, Jean-Michel Krieg, Gary Melnick, Sebastiano Ligori, Boon-Kok Tan, Bertrand Thomas, Volker Tolls, Jeanne Treuttel, Jérôme Valentin; David L. Clements, Ray Blundell, Asantha Cooray, Meredith MacGregor, Ronald J. Vervack, Jr.

***Abstract*—The Heterodyne Spectroscopy Instrument (HSI) is a focal plane array receiver (FPA) for the Far-IR Spectroscopy Space Telescope (FIRSST), a proposal submitted to NASA's APEX call. Though FIRSST was not selected for further study, the receiver design allowed us to quantitatively demonstrate that small (few to few tens of pixels) heterodyne FPAs are now feasible for space missions, that they can be built with high technology readiness level (TRL) components, are reliable and fit within the mass/volume/power constraints of medium to large missions. The design also identified the challenges and critical design considerations, in particular great care needs to be taken to minimize the cryogenic heat load of the cryogenic IF amplifiers and to keep the power consumption of the spectrometer backends low.**

The HSI was designed primarily to study the trail of water from the Interstellar Medium (ISM) to planets and to observe the important low-lying transitions of water and its isotopes between 500 and 2000 GHz undetectable from the ground due to Earth's atmosphere. HSI has a spectral resolving power of up to 10^7 (0.03 km/s), ideal for kinematic studies or line tomography. HSI has six 5-pixel-arrays covering 3 frequency bands and 2 linear polarizations.

The concept study showed that heterodyne array receivers such as HSI can be built today even for fast missions and that they are powerful and versatile instruments, with low-risk technology capable of opening up the THz sky at very high spectral resolving power for new discoveries.

***Index Terms*—Heterodyne instruments, Far-IR space instruments, heterodyne receivers, array receivers, THz astronomy, THz detectors.**

I. INTRODUCTION

TECHNICAL advances have opened up the entire electromagnetic spectrum for astronomy and led to many discoveries and a much better understanding of our Universe. Millimeter and submillimeter observations allowed us to look into dark clouds, probe the interstellar medium, get an inside on how stars and planets form, and how black holes are nourished. Heterodyne receivers have been particularly valuable due to their high spectral resolving power ($R \sim 10^5$ to 10^7), enabling astronomers to measure velocities and hence to observe kinematical processes. Heterodyne receivers have also opened up the field of astrochemistry, determining both molecular and isotopic abundances and the chemical pathways that lead to the molecules found in the ISM and on planets.

Advances in heterodyne receiver technology have pushed the boundaries of astronomical instruments; these advances have led to observations at THz frequencies, have increased the

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sensitivity, have allowed us to obtain spectra over a wide frequency range, have made the construction of the first heterodyne focal plane arrays possible and hence increase the mapping speed of high spectral resolution observations. Every advancement has led to new insights into the Universe. This article describes a design concept for a sensitive, high frequency heterodyne array receiver using the example of the Heterodyne Spectroscopy Instrument (HSI) for Far-IR Spectroscopy Space Telescope (FIRSST), but equally applicable to other missions, such as the Line Emission Terahertz Observatory (LETO) proposed for ESA's M8 call.

II. Heritage and Prior Instruments

Heterodyne receivers have had a long and successful history over the last nearly 100 years, starting with common radios. Following the discovery of the CO (1-0) transition at 115 GHz in the Orion nebular with Schottky mixer technology in 1970 [1], superconducting millimeter and submillimeter heterodyne receivers emerged in the 1980s as well as dedicated (sub)millimeter telescopes such as the Institut de Radioastronomie Millimetrique 30-m telescope [2], the James Clerk Maxwell Telescope [3], and the Caltech Submillimeter Observatory [4]. Others single dish telescopes were added later, such as the Atacama Pathfinder EXperiment [5], the ASTE Telescope [6], and the Large Millimeter Telescope [7]. Heterodyne receivers that allow the measurement of phase and amplitude are also invaluable for connected interferometers, such as the Berkeley Illinois Maryland Array [8], the Owens Valley Radio Observatory [9], the SubMillimeter Array [10] and the Atacama Large Millimeter/Submillimeter Array [11] as well as Very Long Baseline Interferometry, where telescopes around the world are used to observe together and their signals are cross correlated, for example within the Event Horizon Telescope collaboration [12].

Heterodyne receivers also have been used on airborne and space missions. In particular, on airplane observatories, such as Kuiper Airborne Observatory [13] and the Stratospheric Observatory for Far-Infrared Astronomy (SOFIA) [14], e.g. with the GREAT instrument [15]. There have been several heterodyne receivers flown on satellites: on the Odin satellite [16] and on the SWAS satellite [17], and on JUICE [18]. The Heterodyne Instrument for the Far-Infrared (HIFI) [19] operated successfully from 2009 to 2013 on the Herschel Space Observatory [20], one of ESA's cornerstone missions.

Heterodyne receivers observe at high spectral resolving power of 10^5 to 10^7 with 1000 to 10000 frequency channels, but generally with only a single spatial pixel. For astronomical questions that require many spectra over an area of the sky, heterodyne focal plane arrays (FPA) have recently been developed. Most FPAs have few pixels (between 4 and 16). The largest array built so far is Supercam, with 64 pixels [21]. The CHAI receiver, also with 64 pixels, [22] is under construction for the Fred Young telescope. A good overview of the FPAs is given in [23][24].

After successfully operating FPAs on the ground, the first ones have recently been carried by airplanes and balloons. The SOFIA airplane observatory [13] was equipped with the very successful upGREAT array receivers with 2 bands enabling observations of the major cooling lines of [CII] and OH (1.9 THz - 2.5 THz) and [OI] 4.7 THz [25][26]. It had medium-sized arrays of 2 x 7 pixels and 1x7 pixels. On balloons, STO-1 [27] has flown with 2x2 HEB mixer arrays for 1490, 1900 GHz observations; STO-2 [28][29] with two pixel arrays for 1490 and 1900 GHz and recently GUSTO [30][31][32][33] with three 8 pixel arrays for 1.4, 1.9 and 4.7 THz. For atmospheric science the OSAS B balloon-born THz spectrometer for atomic oxygen in the upper atmosphere has carried a 4.7 THz cryogenic heterodyne receiver [34].

No focal plane array has flown on a space mission, yet. There have been studies of heterodyne array receivers for the Millimetron Telescope[35][36][37] and a more detailed study for the Heterodyne Receiver for Origins (HERO) [38] with 4 bands of 9 pixels in two polarizations for the Origins Space Telescope concept study [39][40]. More recent studies included heterodyne array receivers on the SALTUS mission proposal [41][42]. The Far Infrared Spectroscopy Space Telescope proposal allowed us to carry out a new heterodyne array design for a smaller telescope, confirming its great astronomical interest on one hand, and its feasibility for medium size missions on the other.

III. Far-IR Spectroscopy Space Telescope

The Far-Infrared Spectroscopy Space Telescope (FIRSST) was one of three far-IR proposals submitted to the NASA Astrophysics Probe Explorer (APEX) call in November 2023, a mission with a budget between a medium and large mission and a fast turn-around time aiming for a launch in 2032.

FIRSST was designed around three fundamental science questions: 1) How do planets form in disks and what causes their diversity? 2) What is the source of water in planet forming disks and how do oceans accumulate water? 3) How does the intergalactic medium influence star formation and how do galaxies grow?

To answer these questions the FIRSST mission had a 1.8m cryogenic mirror cooled to ~4.8K with a closed cycle cryocooler. (The cooled mirror is essential only for the incoherent receiver of FIRSST as it lower its infrared background. A heterodyne receiver can tolerate higher mirror temperatures as it has much smaller spectral bandwidth. A mirror at 4.8K with an emissivity of 2% adds 0.1K to the heterodyne receiver noise temperature, one at 300K 6K. Both numbers are negligible compared to system noise temperature.) FIRSST had an enclosed architecture for stability, to minimize stray light and to allow to point anywhere in more than half the sky (~54 %) at any moment. FIRSST was to carry two scientific instruments: the Direct Detection Spectrometer Instrument (DDSI) using microwave Kinetic Inductance Detectors (mKIDs) (PI Gordon Stacey and Karwen Rostem) and the Heterodyne Spectroscopy Instrument (HSI), described in detail in this paper. FIRSST was to observe from L2 and had a minimum mission lifetime of 5 years.

IV. Requirements and Constraints for the Heterodyne Spectroscopy Instrument

A. Scientific Motivation and observing requirements

A key science driver for HSI/FIRSST was the trail of water

from the interstellar medium to rocky planets. Even though, FIRSST was not selected, the scientific questions remain and will hopefully be addressed by another mission. As liquid water is believed to be necessary to develop any form of life, possible pathways of how water can be captured by rocky planets are of great interest. HSI/FIRSST was to determine whether water in planet-forming disks is inherited from the Inter Stellar Medium (ISM) or regenerated by observing the ortho-para ratio of water and the HDO to H_2O ratio. Water in pre-stellar cores and protoplanetary disks is of particular interest as it has hardly been observed before. In addition, HSI/FIRSST was to observe comets in order to investigate how inner planets including Earth received water. In particular, HSI/FIRSST was to observe the emission lines of H_2O and HDO and map the D/H ratio in the coma of periodic comets and Oort cloud comets.

Gaseous water is traced by its emission and absorption lines [43]. Most of the water in a planet forming disk is cold, some will be frozen out and its broadband signatures can be observed in the mid- to far-infrared. Gaseous water is best traced by its low-lying transitions (at 557, 1661, and 1670 GHz for ortho-$H_2{}^{16}O$; 1113 and 988 GHz for para- $H_2{}^{16}O$; 548 GHz for $H_2{}^{18}O$; 552 GHz for $H_2{}^{17}O$; and 509 and 894 GHz for HDO).

Planet forming disks are very small with scales ranging from ~70 to ~280 AU [44] and cannot be resolved with a single dish far-IR telescope. However, observations at high spectral resolving power ($R > 10^6$, < 0.3 km/s) allow us to use line tomography to determine the gas distributions. As the rotation speed in the outer part of the disk is slower than in the inner part, the Doppler shift of the lines emitted by gas with slower motions from the outer part of the disk is smaller than the one from the inner part. Using the velocity variations of a rotating disk, the observed spectrum can be inverted into a gas distribution. High spectral resolution ($R = 10^5$ to10^7) is not only necessary for line tomography but also to determine gas kinematics and to distinguish between absorption versus emission features.

B. Science Traceability and Technical Requirements

From the science case described above a science traceability matrix was deduced with the following requirements for the performance of the heterodyne receiver:

1) Coverage of the most important water lines and its isotopes at 509, 557, 1113, 1229, 1278 and 1670 GHz as well as the [CII] emission line at 1900 GHz.
2) A sensitivity of $2.5x10^{-20}$, $2.1x10^{-19}$ and $2x10^{-19}$ Wm^{-2} for bands 1,2 and 3 at 5 sigma in 10 h integration at 0.1 km/s resolution. For a 1.8 m telescope and an aperture efficiency of 0.8, this translates into double sideband receiver noise temperatures of 80 K, 500 K and 525 K, respectively.
3) Angular resolution smaller than 90".

HSI SCIENTIFIC REQUIREMENTS			
Parameters	Band 1	Band 2	Band 3
Wavelength range (μm)	600-380	340-240	200-150
Frequency range (GHz)	500-790	882-1250	1500-2000
Sensitivity (Wm^{-2})	$2.5x10^{-20}$	$2.1x10^{-19}$	$2x10^{-19}$

Table 1: Scientific Requirements

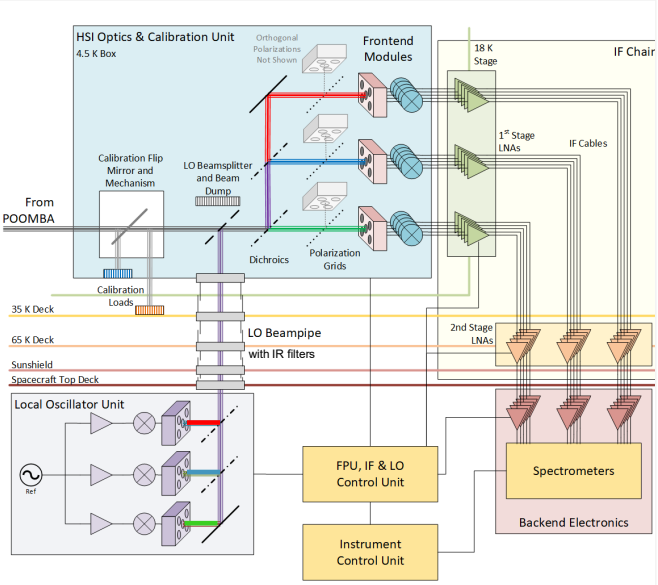


Figure 1: HSI architecture follows the basic principle of successful heterodyne instruments such as HIFI/Herschel. HSI has 3 frequency bands, and in both polarizations a 5-pixel array, allowing for mapping speeds that are orders of magnitude faster than HIFI. (Image Credit: Paul Grimes, CfA.)

4) Very high spectral resolving power $R = \lambda/\Delta\lambda$ of better than $3x10^6$, i.e., better than 0.1 km/s.
5) Arrays of at least 2 pixels for simultaneous on/off integration.
6) An instantaneous frequency bandwidth of at least 4 GHz.

In addition, the instrument had to fit within the spacecraft (<1 m^3, <100 kg), require little electrical power (<300 W), and most importantly exert a small heat load (< 30 mW) on the 4.5 K cryocooler.

V. Conceptual Design of the Heterodyne Spectroscopy Instrument

The Heterodyne Spectrometer Instrument (HSI) followed the basic design of any heterodyne receiver, where the signal from the sky is mixed with an artificial, monochromatic reference signal from the Local Oscillator (LO) in order to create a beat signal at a lower Intermediate Frequency (IF) that can be more easily amplified and spectrally detected (see Figure 1).

HSI builds strongly on the heritage of ALMA [11], HIFI/Herschel [19], upGREAT [26] and the design of HERO/Origins [38]. HSI has, however, special features: it was designed to be the first heterodyne array receiver on a space mission. HSI has a large frequency coverage, focal plane arrays, multi-frequency observing modes, it is versatile, but low risk and fits within the mass, volume, power and cooling budget of a medium size spacecraft. HSI cooling budget per pixel is a factor of ~5 to ~10 lower than that of HIFI/Herschel and a factor of ~10 to ~100 lower than that of ground based heterodyne receivers. HSI covers a nearly continuous frequency range 500 to 2000 GHz in only 3 bands. Power, volume and mass are also reduced with respect to HIFI/Herschel while providing focal plane arrays.

Figure 1 displays a schematic diagram of HSI: the light from the sky is entering on the left after being collected by FIRSST's primary mirror and redirected by the Pick-Off Optics and Mirror Beam-Steering Assembly (POOMBA). The Local Oscillator Unit, located in the warm space bus, synthesizes three LO signals that are superimposed onto each other and then optically guided through the LO beampipe with several heat-sunk infrared filters to the cold frontend module, where they are superimposed onto the beam from the sky. The combined signal is then divided into three frequency bands using dichroics. Each frequency band is further split into its two linear polarizations. Each of the 3 bands x 2 polarizations is then imaged onto a 5-pixel array (Frontend Module), where it is mixed to an intermediate frequency (IF). The IF is amplified in three stages at 18, 65 and 300 K and finally spectrally detected. The Instrument Control Unit provides communication between the main on-board computer and the receiver, where subunits are commanding the Focal Plane Array, the IF and the LO.

VI. Components of the Heterodyne Spectroscopy Instrument

The components for HSI need to be selected to have optimal performance, but also to require little power, volume, mass and cooling resources, and to have high technology readiness level (TRL) (ideally ≥ 6) to allow a launch within 8 years.

One crucial parameter is the sensitivity of the receiver, which is generally quoted as receiver noise temperature. The system noise temperature includes the receiver but also all the optics, including the primary mirror. The receiver noise temperature can be calculated from the noise temperatures T_{Ci} and the gains G_{Ci} of its individual components Ci as $T_{rx}=T_{C1}+T_{C2}/G_{C1} + T_{C3}/(G_{C1}xG_{C2}) + \ldots$ The receiver noise temperature is dominated by the first components up to the first amplifier. For FIRSST the mixer temperature contributes most to the receiver noise temperature as well as the losses in the receiver optics. The rms noise (in K) of an astronomical observation is $T_{sys}/\mathrm{sqrt}(\Delta\tau*\Delta\nu)$, where $\Delta\tau$ is the integration time and $\Delta\nu$ is the spectral resolution. FIRSST uses double sideband (DSB) mixers, therefore we quote DSB noise temperatures. However, spectral line (in contrast to continuum)observations will only fall into 1 sideband and the single sideband (SSB) noise is approximately twice as high.

1) Mixer

HSI uses the most sensitive superconducting mixers available: Superconducting-Isolating-Superconducting (SIS) mixers and Hot Electron Bolometer (HEB) mixers. SIS mixers provide near quantum noise limited performance at frequencies up to ~1 THz due to the superconducting band gap, e.g. [45] [46]. State-of-the-art SIS mixer noise temperatures at 600 GHz are 25 K [47].

HEB mixers work well up to several THz, e.g., 4.7 THz in upGREAT[19]. Receivers with HEB mixers have receiver noise temperatures > 4k/hv and their IF bandwidth is below 7 GHz [48]. There has been, however, significant progress since the first HEB receivers [49][50] and the HIFI/Herschel instrument [19] and DSB receiver noise temperatures have been significantly reduced to 450 K at 1.3 THz [51] 530 K at 1.6 THz, 640 K at 2.5 THz and 2200 K at 5.3 THz[52].

Fig. 2 shows published SIS and HEB DSB receiver noise temperatures as well as derived DSB mixer noise temperatures. The mixer noise temperatures cannot be measured directly, but are derived from the receiver noise temperatures (or system noise temperatures) using the estimated optical losses and the measured IF noise temperature. The FIRSST minimum expected and goal DSB

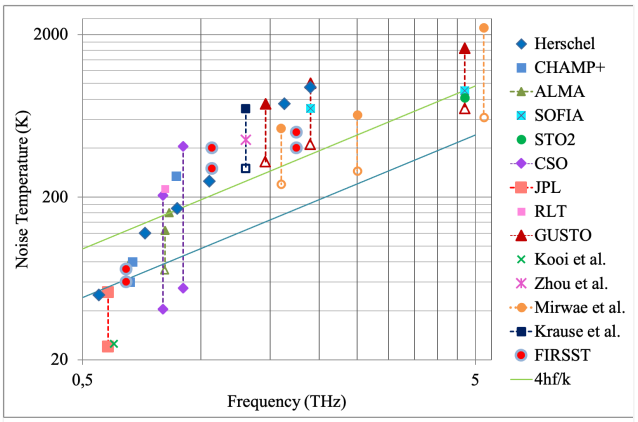


Figure 2: DSB Receiver (solid symbols) and mixer (empty symbols) noise temperatures of different receivers. The FIRSST current best estimate (CBE) and the Minimum expected value (MEV) DSB receiver temperatures are indicated by the large red circles.

receiver noise temperatures are marked with large red and blue circles and lie roughly within the published results. However, noise temperatures depend not only on the RF frequency, but also on the IF frequency and are sometimes quoted as the best value, sometimes as the average value over a certain IF and/or RF range.

FIRSST/HSI Band 1 (500 to 790 GHz) uses SIS mixers similar to those of HIFI/Herschel Band1 [53], but with slightly wider RF bandwidth. We expected to have similar receiver and mixer noise temperatures as HIFI/Herschel, however FIRSST mixers are warmer (4.7 K versus 2 K), so reaching this goal temperature still needs small improvements. For Band 2 (882 to 1250 GHz) we are pursuing two alternatives: HEB mixers can easily operate at those frequencies, have sufficient IF bandwidth and are high TRL. HSI requires lower receiver noise temperatures than has been reached by the HIFI HEB channel; it requires T_{rx} comparable to those published by [33]. (Using the best published mixer noise temperature of 330 K at 1.46 THz [33] and a mixer loss of 5.7 dB, optical losses of 0.83 dB, optical noise of 3.8 K and an IF noise of 4 K, an instrument with the above HEBs would have a DSB receiver noise temperature of 421 K at 1.46 THz.) Alternatively, SIS mixers will have lower noise and are expected to be more stable, but no high TRL versions are available today and we need to carry out technical development and space qualifications. Band 3 (1500 to 2000 GHz) uses HEB mixers. The receiver noise level we require is slight lower than has been reached by prior instruments.

2) Local Oscillator

HSI uses amplifier-multiplier chains as local oscillators; where a signal between typically 10 and 20 GHz from a

synthesizer is multiplied up by a chain of doublers and triplers and enhanced by amplifiers at W or F bands. Quantum Cascade Lasers (QCL) [54][55][56][57][58][59] produce more power above 1 THz and have been successfully operated on upGREAT/SOFIA[60], but they require cooling to below 77K and produce a significant heat load, which is the principle reason to not use them on FIRSST.

HSI exploits the significant progress made with Schottky diode multipliers [61][62][63][64][65][66]. HSI's LOs build on the heritage of the space-qualified LOs for the Submillimeter Wave Instrument (SWI) [68]of the Jupiter Icy Moon Explorer (JUICE) [18]. This TRL8 LO gives 2 mW of output power with a power consumption of 8 W (3 W amplifier, 5 W synthesizer) at 600GHz [68]. Siles et al. [69] have demonstrated 700 µW of output power at 1.6 THz using high-power GaN amplifiers with an output power of 2 W, assuming a power added efficiency of 10% [70], this requires 20 W power consumption.

The SIS as well as the HEB mixers require 100 to 500 nW [71] on the mixer element and between 500 nW and 1000 nW at the entrance of the cryostat/mixer unit. If we add a 10 dB beam combiner, we would require 5 to 10 µW per pixel, 25 to 50 µW for 5 pixels, but should add a margin of at least 20 % for losses in the LO power coupling, i.e. 30 to 60 µW per polarization or 60 to 120 µW for both polarizations of each band. This power can be provided by current LO chains, but care needs to be taken that the LO covers a large enough RF bandwidth, or alternatively several LO chains are required to cover one full band.

3) *Low Noise Amplifiers*

The first amplifier in the Intermediate Frequency (IF) chain is critical as its noise can easily deteriorate the overall receiver sensitivity. Two types of cryogenic low noise amplifiers were considered for HSI: either SiGe LNAs based on HBT (heterojunction bipolar transistor) or InP High Electron Mobility Transistors (HEMT) amplifiers. SiGe LNAs exert very little heat load (as low as 0.3 mW), but often have narrow IF bandwidth (e.g., 1.8 GHz [72], 4 GHz [73]). Alternatively, InP HEMTs [74][75] have wide bandwidth (up to 2 octaves). The HIFI/Herschel InP have very low noise (< 4 K), wide bandwidth (4GHz), high gain (27 dB), and are at TRL (~8). However, the HIFI/Herschel InP HEMTs produce a heat load of ~5 mW, which would limit HSI to operate at most 5 amplifiers / pixels at any time. Several approaches are used to reduce the heat load: First of all, development is ongoing to reduce the driving voltage and current of InP amplifiers. Secondly, we are experimenting to reduce the applied voltage to below the nominal voltage, which reduces the gain and increases the noise slightly [76][77]. If the stability is not affected a compromise between lower heat load and less gain can be found. Thirdly, HSI places the amplifiers on the 18 K stage where a higher heat load can be absorbed. The amplifiers are positioned physically close to the mixer in order to minimize any losses in the connecting cable.

Parametric amplifiers [78][79] are a very exciting alternative for future missions, as they have extremely low heat dissipation, but they do not have the required TRL level for FIRSST, yet.

4) *Spectrometer Backends*

HSI uses two types of backends both with heritage from SWI/JUICE. The Chirp Transform Spectrometers cover 1 GHz bandwidth with 10000 channels, ideally suited for very high spectral resolution observations. The Autocorrelator Spectrometers (ACS)[80][81] have a bandwidth of 4 GHz covered by 1024 channels. Both backends have relatively low power consumption of 7 W. An alternative, would be spectrometers that use Field Programmable Gate Arrays (FPGAs), but currently no space qualified low power chips are available.

VII. Optical and Mechanical Design

Figure 3 shows the optical and mechanical design of the Heterodyne Spectrometer Instrument. The optics and the mixers are located on the 4.8 K optical bench just below the primary mirror (at the top of the figure), while the LO unit is located in the warm space bus (at bottom). In Cold Optics Assembly (see zoom on right) the telescope beam enters from the right (from the POOMBA) and is refocused. In the Beam Combination Module, the LO beams are superimposed onto the signal from the sky. It also contains a cold and a warm blackbody calibration load that can be moved into the line of sight with a switch mirror. The Beam Combination Module contains the dichroics that redirect the light of the respective wavelength to the different mixer modules. Each mixer module splits the light in its two linear polarizations and refocuses it on the 5-pixel focal plane arrays. The LO unit in the warm space craft, contains 6 Amplifier Multiplier Chains (1 active, 1 spare) for each of the 3 bands. The LO signals are superimposed onto a single beam, refocused to a slow beam and sent to the Cold

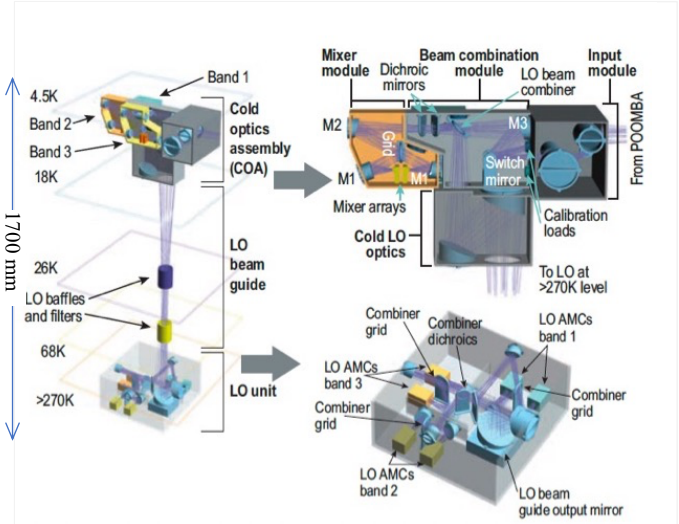


Figure 3: Optical and mechanical design of HSI/FIRSST. Optics, mixers and first LNA reside on the cold optics bench, while the LO Unit is in the warm space bus. The design is compact and modular. (Image Credit: Andrey Baryshev, Groningen.)

Optics Assembly via a single LO beam guide. The beam guide contains baffles and filters to minimize the thermal heat load and any stray light. The Cold LO Optics receives the beam and refocuses it before it sents on to the Beam Combination module.

The optics design is high TRL. The mirrors are made of aluminum with an accuracy of ~1 µm.

Optics losses in the HSI signal chain are due to LO beam splitter loss of 10 %, and the dichroic filters. For the latter we assume a reflective loss of 2.5 % and a transmission loss of 5 %, all located to 18 K temperature. For additional losses of each of the 8 mirrors we assume 0.3%, for the polarization grid 1% and for the feed horns also 1%. This results in 19.4 % (0,94dB) loss for band 1, 17.3 % (0.83dB) for band 2 and 17.4 % (0.83dB) for band 3. For an edge taper of 11dB, we expect an aperture efficiency of 80%. These losses are included in the system budget.

There are some challenges: A single LO needs to pump all 5 (or 10) mixers of the focal plane array. We intend to split the LOs in waveguide at the output of the LO, see e.g., [82]. An alternative would be Fourier Gratings or Phase Arrays [83][84] that have been demonstrated e.g., on GREAT. However, the Fourier Gratings are usually limited to ~5 % RF bandwidth [85].

If the mechanical structure between the space bus and the cold optics deforms upon cool down, the beam might be misaligned and a tip-tilt beam compensation mechanism might be needed on the last LO mirror to ensure that the LO beam is correctly transmitted through the beam guide to the cold optics assembly.

VIII. Instrument Performance

HSI PARAMETERS				
Parameter		**Band**		
		Band 1	Band 2	Band 3
Wavelength (μm)		380 - 600	240 - 340	150 - 200
Frequency (GHz)		790 - 500	1250-882	2000-1500
Resolving power (λ / Δλ)		10^6 to 10^7		
Beam size		52”-83”	33”-47”	21”-28”
Instantaneous FoV		300”x200”	150”x100”	150”x100”
Spectral channels		1024 or 10000		
Array size		5 pixels x 2 polarizations		
Aperature efficiency		80%		
Mixer Type		SIS	HEB	HEB
Receiver noise temperature (DSB)	CBE	60K	300K	400K
	MEV	72K	400K	500K
	Sci Reqt.	80K	430K	525K
IF bandwidth		4 GHz		
Optical bench temperature		4.7K with ± 0.1K stability (not critical)		
LNA temperature (1st stage)		18K with ± 0.1K stability during Allan time		
Mixer temperature		4.7K with ± 10mK stability during Allan time		
RMS WFE budget (nm)	Required	<7500		
	Allocated	3000		
	Margin	250%		

Table 2: HSI parameters and expected performance.

Table 2 summarizes the performance of the Heterodyne Spectroscopy Instrument for FIRSST.

Most of the HSI instrument design and components are well proven and high TRL, and have successfully flown on several space mission, including HIFI/Herschel. However, HSI goes beyond HIFI, where the main advances are: 1) wider RF bandwidths, 2) lower receiver noise and 3) having 5-pixel arrays.

HSI has 45 %, 35 % and 29 % relative bandwidth, which is easily attainable by the SIS and HEB mixers and the challenge lies in providing enough LO power over the whole band. As seen in the LO section, recent LO output power at 1.6 THz is a factor of > 5 higher than required, so a compromise for larger bandwidth should be possible. In worst case more than one LO amplifier or LO chain have to be used for a single band.

The DSB receiver noise temperatures are discussed in the mixer section. They are challenging, and some improvement (~20 %) from the current state is required for band 2 and 3.

The simplest 5-pixel arrays consist of 5 individual high TRL pixels. In the component section it was shown that the heat load of the amplifies can be reduced, and that low power spectrometer backends and LOs exist, so that it is possible to keep the operation of two 5-pixel arrays within a reasonable heat and power budget.

There is no minimum requirement on the Allan variance and Allan minimum times of 10 to 20 sec for SIS mixers and 5 sec for HEB are sufficient.

IX. Volume, Mass, Power and Heat Load Budget

Volume, Mass, Power and Heat Load				
Subsystem	**Location**	**Mass**	**Volume**	**Power**
LO	300 K	17 kg	30 dm³	26 W
Frontend	4.5K Optics Bench	15 kg	44 dm³	14 mW@4.7 K
IF	18-65K-300K	18 kg	10 dm³	24 mW @18 K 0.6 W @65 K 5 W @300 K
Spectro-meter	300K Bus	8 kg	10 dm³	84 W
Control Electronics	300K Bus	17 kg	12 dm³	53 W
Total		**75 kg**	**106 dm³**	**194 W**

Table 3 HSI parameters and expected performance.

Table 3 gives an overview of the mass, volume and power budget. The mass and the volume are dominated by the cold and LO optics and their support structure, as well as the electronics boxes.

The power is dominated by the spectrometers and the control electronics. The spectrometers use 7 W per spectrometer. Assuming only 1 of the two kinds of spectrometers is used, and the observations are carried out with the maximum of 12 pixels this requires 84 W. The control electronics includes mixer bias supplies, calibration unit control, IF bias supply, temperature monitors, bias for LO, a warm IF unit controller, DC/DC converter cards, and are estimated to require 53 W. The LO power is also critical. The JUICE synthesizer requires 5 W, the amplifier in the LO chain 3 W and can easily pump the two 5-pixels arrays. The 1.6 THz LO chain by Siles et al. [69][70], use 20 W for 700 μW output. HSI requires only 60 μW for 5 pixels in one polarization or 120 μW for both polarizations, i.e., ~1/5. Assuming a linear correlation, this suggests ~4 W for the amplifiers, slightly less for band 2 that is at lower frequencies and slightly more for band 3. When all 3 bands are observing simultaneously the 3 LOs require 26 W.

Cryogenic cooling in space is difficult. It can be achieved with liquid cryogenic gases (He, N) that cool by evaporation, as in the case of Herschel, however, this limits the lifetime of the satellite. An alternative are space qualified closed-cycle coolers. They require hundreds of Watts and deliver cooling powers of several tens of mW in the 4 K stage, and around

100 mW in the next stage around 18 K, e.g., ESA promises for the M8 missions 30 mW of heat lift at 4.5 K, 100 mW at 20 K and 500 mW at 120 K with a 500 W input [86], BAE offers 54 mW at 4.5 K, 103 mW at 18 K and 4900 mW at 65 K [87]. Therefore, it is essential to limit both the passive and active heat dissipation in the cryostat. The passive heat dissipation is due to mechanical support and cables, that can be lengthened to minimize heat transfer (except when electrical losses need to be minimized, such as for the IF cables between mixer and first amplifier), and incident radiation, which can be limited by adding heat shields. For FIRSST the LO beam guide from 4 to 300K is equipped with several infrared filters that are heat sunk such that a heat load of an only ~1 mW at 4 K was calculated, as well as an additional 2mW from the hot calibration load. The active heat load is dominated by the IF amplifiers with 2 mW per amplifier. For FIRSST the amplifiers are at the 18 K stage; with a maximum of 12 amplifiers running simultaneously, the heat load is 24 mW. Additional heat loads come from any motors in the cold optics assembly, for FIRSST the calibration switch mirror with a dissipation of around 10 mW. The mixers have negligible heat load of ~1 µW. At 65 K we estimate a heat load of 12x50 mW i.e., 600 mW from the second stage amplifiers. At ambient temperature an upper limit for the heat load is the components power consumption.

X. Observing modes

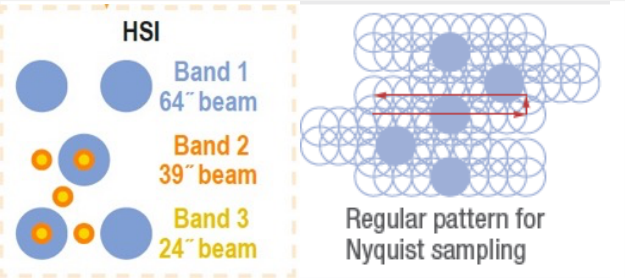


Figure 4: Left: Footprint of the three Focal Plane Arrays of HSI. Right: On the fly pattern for efficient mapping. (Image Credit: Ron Vavreck, APL)

Figure 4 shows the footprint of the 3 bands of HSI, where the two linear polarizations are superimposed.

There are two major observing modes: one for pointed observations of compact or not resolved sources, the other for mapping of extended sources. For pointed observations a clever scheme is used where the two pixels of the array that contain all the 3 bands are used, so that one is on source, while the other off source. In the next step the source is moved to the former off-source pixel, i.e., the source is double pixel switched. This allows the source to be always in one of the two pixels and to decrease the observing time by a factor of 2 compared to the single pixel on/off observations. The pointed observations can be carried out in dual-polarization and triple-frequencies, so that in total 2x3 = 6 on-source and 6 off-source spectra are obtained simultaneously.

For mapping a first snapshot can be obtained with a single pointing. To Nyquist sample a small area the array will be dithered, filling in the missing beams. For larger maps, on-the-fly mapping is proposed as indicated on the right of Figure 4. In order to increase observing efficiency a single on-the-fly map will have a maximum size of ~12 arcmin, which corresponds to the throw of the beam steering mirror, a movable tertiary mirror. Several of these maps can then be mosaicked together for larger maps. HSI uses dual polarization mapping to increase its sensitivity. The different frequency bands have to be mapped sequentially, because the beams have different sizes and the power consumption of 30 backends is not supported in the current design, nor the heat load of 30 amplifiers. However, 10 pixels can be observed simultaneously.

XI. Conclusion

The Heterodyne Spectrometer Instrument builds strongly on the successful HIFI/Herschel instrument, on upGREAT/SOFIA, and the HERO/Origins study, as well as on other balloon and ground-based missions. To adopt to a space mission, special care was taken to minimize the heat load, as well as the required power, mass and volume. Only high TRL components were used to allow for a quick launch in 2032.

The quantitative design confirmed that small (a few to a few tens of pixels) heterodyne focal plane arrays are now a feasible and a highly reliable option even for medium size space mission. HSI is a useful concept onto which future heterodyne focal plane arrays can be based. These heterodyne cameras will be the key to high spectral resolution observations in the far-IR.

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